Lecture 4: Straight Line Complexity of Linear Programs

Algorithms & Geometry of Linear Programming

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Outline

From simplex to interior point methods

Lectures 2–3: the simplex method — exact, vertex-walking; its practical success explained probabilistically.

Today: interior point methods (IPMs).

- *approximate* methods: drive down the optimality gap $c^\mathsf{T} x - v^*$
- *how many iterations to halve the gap?*
- *how many iterations to optimal solution?*
  (reducing gap to $2^{-\Theta(L)}$ is sufficient, $L =$ bit size)

Iteration bounds for interior point methods

IPMs follow the central path: as far from the boundary as the objective allows. Standard form, $\mathbf{A} \in \mathbb{R}^{m \times n}$, feasible region of dimension $d = n - m$; one iteration $=$ $O(1)$ linear system solves.

- **[Renegar '88]** log-barrier path following: gap halved every $O(\sqrt{n})$ iterations; exact optimum after $O(\sqrt{n}\,L)$
- **[Nesterov–Nemirovskii '94]** universal barrier: $O(\sqrt{d})$ per halving — hard to evaluate (volumes); computable via Lewis weights: $\tilde{O}(\sqrt{d})$ **[Lee–Sidford '14]**
- **[Vavasis–Ye '96]** layered least squares follows the *entire* path in $O\big(n^{3.5}\log(\kappa_{\mathbf{A}}\, n)\big)$ iterations, $\kappa_{\mathbf{A}}$ circuit imbalance — no dependence on b, c
- **[ABGJ '18]** lower bound: instances where path following takes $2^{\Omega(n)}$ iterations

Which of these bounds reflects reality?

Two phenomena the bounds do not explain

Phenomenon 1: practice beats theory

Solvers routinely converge in 20–50 iterations, essentially *regardless of dimension* — far better than $O(\sqrt{n}\,\log(1/\varepsilon))$.

Phenomenon 2: no strong polynomiality [ABGJ '18]

There are instances with $O(n)$ constraints on which *every* path-following IPM needs $2^{\Omega(n)}$ iterations.

- how can a method with a proven $O(\sqrt{n}\,L)$ bound need exponentially many iterations?
- what distinguishes easy instances from hard ones?

The question, reformulated

Fix only the paradigm — *follow the central path* — and ask:

How many iterations does the geometry of the path itself allow?

The answer is a concrete quantity: the straight line complexity (SLC) of the linear program — a property of the *instance*, not of any particular method or barrier.

1. **Block 1:** the SLC framework — a lower bound for *every* path-following method, nearly matched by the method of Lecture 5 [ADLNV '22]
2. **Block 2a:** SLC can be exponential [ABGJ '18] — via tropical geometry
3. **Block 2b:** SLC $\leq O\big(\min\{m, n-m\} \log \kappa_{\mathbf{A}}\big)$, with $\kappa_{\mathbf{A}}$ the circuit imbalance [DKNOV '24]

Setup: the standard-form pair

(P) $\min c^{\mathsf{T}} x$ s.t. $\mathbf{A}x = b,\ x \geq \mathbf{0}$ (D) $\max b^{\mathsf{T}} y$ s.t. $\mathbf{A}^{\mathsf{T}} y + s = c,\ s \geq \mathbf{0}$

- $\mathbf{A} \in \mathbb{R}^{m\times n}$ full row rank; both sides strictly feasible; $v^* :=$ common optimal value; (x^*, s^*, y^*) optimal
- **gap formula** (Lecture 1): for feasible x, (s, y): $c^{\mathsf{T}} x - b^{\mathsf{T}} y = s^{\mathsf{T}} x \geq 0$
- in particular $s^{*\mathsf{T}} x = c^{\mathsf{T}} x - v^*$: a nonnegative functional vanishing exactly on the optimal face — we write the gap in this form from now on
- **gap-truncated regions** (bounded, by dual strict feasibility):

$$\mathcal{P}_g := \{x : \mathbf{A}x = b,\ x \geq \mathbf{0},\ s^{*\mathsf{T}} x \leq g\}, \qquad g \geq 0$$

Outline

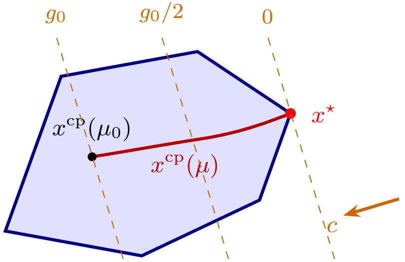
The central path

Definition (Central path)

For $\mu > 0$:

$$x^{\mathrm{cp}}(\mu) := \underset{\mathbf{A}x=b,\ x>\mathbf{0}}{\operatorname{argmin}}\ c^{\mathsf{T}}x - \mu\sum_{i=1}^{n}\ln x_i$$

- the log-barrier blows up at the boundary: interior minimizer, unique
- trades off distance to the boundary against progress toward the optimum
- μ large: near the "center"; $\mu \to 0$: $x^{\mathrm{cp}}(\mu) \to$ optimal $x^\star$
- a path-following IPM tracks the path as $\mu \to 0$ (Lecture 5)

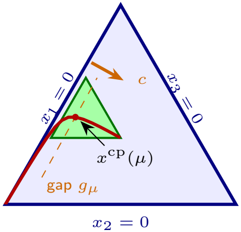


The path nearly maximizes every coordinate

Lemma

Let $\mu > 0$ *and let* $x' \in \mathcal{P}$ *satisfy* $c^\mathsf{T} x' \le c^\mathsf{T} x^{\mathrm{cp}}(\mu)$. *Then* $x'_i \le n\, x_i^{\mathrm{cp}}(\mu)$ *for every* $i \in [n]$.

Equivalently: $x^{\mathrm{cp}}(\mu)$ lies in the shaded region — every coordinate at least a $\frac{1}{n}$-fraction of its maximum over $\mathcal{P}_{g_\mu}$, $g_\mu :=$ gap of $x^{\mathrm{cp}}(\mu)$.

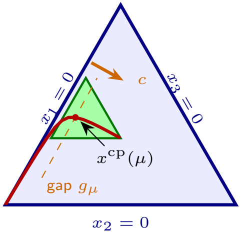


shaded: $x_i \ge \frac{1}{3} \max\{x'_i : x' \in \mathcal{P}_{g_\mu}\}$ for all i

The path nearly maximizes every coordinate

Lemma

Let $\mu > 0$ and let $x' \in \mathcal{P}$ satisfy $c^\mathsf{T} x' \le c^\mathsf{T} x^{\mathrm{cp}}(\mu)$. Then $x'_i \le n\, x^{\mathrm{cp}}_i(\mu)$ for every $i \in [n]$.



shaded: $x_i \ge \frac{1}{3} \max\{x'_i : x' \in \mathcal{P}_{g_\mu}\}$ for all i

Proof. Write $x := x^{\mathrm{cp}}(\mu)$: an interior minimizer of $f(w) := c^\mathsf{T} w - \mu \sum_i \ln w_i$ over $\{w : \mathbf{A}w = b\}$, so $\nabla f(x) = c - \mu\, x^{-1} \perp \ker(\mathbf{A}) \ni x' - x$. Hence

$$0 \;=\; (c - \mu\, x^{-1})^\mathsf{T} (x' - x) \;=\; \underbrace{c^\mathsf{T} x' - c^\mathsf{T} x}_{\le 0} \;-\; \mu \sum_{i=1}^{n} \frac{x'_i}{x_i} \;+\; \mu\, n \quad\Longrightarrow\quad \sum_{i=1}^{n} \frac{x'_i}{x_i} \;\le\; n.$$

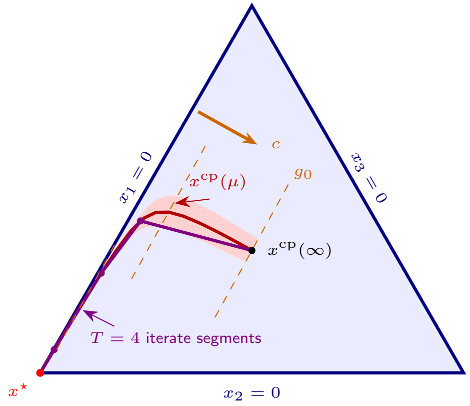
Every term is nonnegative, so each single term is at most n. $\square$

Path-following methods, abstractly

Model for Path-following IPMs:

- iterates stay multiplicatively close to the path: every *slack* within a factor γ of the path point with the same gap (standard form: the coordinates x_i)
- consecutive iterates joined by segments inside this multiplicative tube
- $\Rightarrow$ a T-iteration run traces a piecewise linear curve with T pieces inside the tube

Lower bounding T is now geometric: *how few pieces can such a curve have?*



$\min x_1 + \varepsilon x_2 + \varepsilon^2 x_3$ over the simplex, $\varepsilon = \frac{1}{50}$; tube factor 1.1, computed exactly

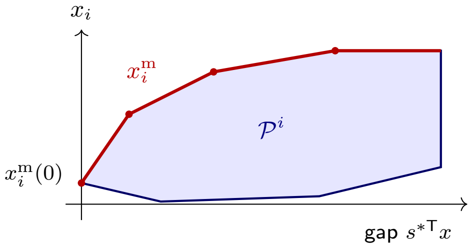
Outline

The max central path

Definition (Max central path)

For $i \in [n]$ and $g \geq 0$: $x_i^{\mathrm{m}}(g) := \max\{x_i : x \in \mathcal{P}_g\}$, the largest i-th coordinate at gap g.

- each x_i^{m} is nondecreasing, concave, piecewise linear
- the vector $x^{\mathrm{m}}(g)$ is generally *not* feasible — one concave curve per coordinate



Sandwiching the central path

Write $g_\mu := s^{*\mathsf{T}} x^{\mathrm{cp}}(\mu)$ for the gap of the central path point $x^{\mathrm{cp}}(\mu)$ (an increasing function of μ).

Corollary (Sandwich)

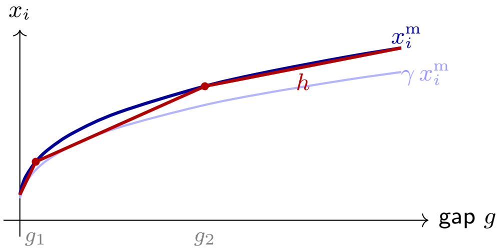
$$\frac{1}{n}\, x_i^{\mathrm{m}}(g_\mu) \;\leq\; x_i^{\mathrm{cp}}(\mu) \;\leq\; x_i^{\mathrm{m}}(g_\mu) \qquad \text{for all } \mu > 0,\; i \in [n]$$

Upper bound: $x^{\mathrm{cp}}(\mu) \in \mathcal{P}_{g_\mu}$. Lower bound: the Lemma, applied to the maximizer attaining $x_i^{\mathrm{m}}(g_\mu)$.

Straight line complexity

Definition (Straight line complexity)

For $f : [0, g_0] \to \mathbb{R}_+$ and $\gamma \in (0,1]$, $\mathrm{SLC}_\gamma(f, g_0) :=$ minimum number of pieces of a continuous piecewise linear $h : [0, g_0] \to \mathbb{R}_+$ with $\gamma f \le h \le f$ on $[0, g_0]$.



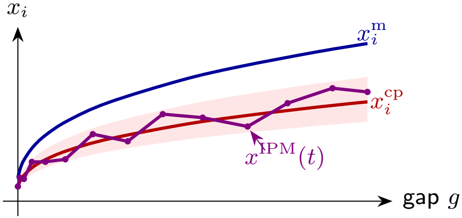
Computed: $x_i^{\mathrm{m}}(g) = 1.6(g + 0.03)^{0.35}$, $\gamma = 0.86$; the greedy h (maximal chords) has 3 pieces and is optimal, so $\mathrm{SLC}_\gamma(x_i^{\mathrm{m}}, g_0) = 3$.

SLC lower-bounds every path-following method

Proposition

Let $x^{\mathrm{IPM}}(t)$ be a piecewise linear curve in $\mathcal{P}$ with T pieces, its gap strictly decreasing from g_0 to 0, satisfying the tube condition $\gamma\, x_i^{\mathrm{cp}} \le x_i^{\mathrm{IPM}}(t) \le \gamma^{-1} x_i^{\mathrm{cp}}$ (at matching gaps, for all i). Then

$$T \;\ge\; \max_{i \in [n]} \mathrm{SLC}_{\gamma^2/n}\big(x_i^{\mathrm{m}}, g_0\big).$$



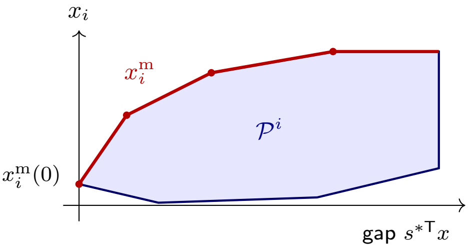
Each iterate segment of x^{IPM} is one linear piece of x_i as a function of the gap; tube (shaded) + sandwich place $\gamma\, x_i^{\mathrm{IPM}}$ in $[\frac{\gamma^2}{n} x_i^{\mathrm{m}},\, x_i^{\mathrm{m}}]$. Here $T = 13$; 3 pieces would do.

The max central path is a shadow

Project $\mathcal{P}$ onto the plane of the gap and coordinate i:

$$\mathcal{P}^i := \{ \left(s^{*\mathsf{T}} x,\ x_i \right) : x \in \mathcal{P} \}$$

- graph of x_i^{m} = increasing part of the upper boundary of $\mathcal{P}^i$
- breakpoints = projected vertices, pieces = projected edges: a shadow vertex simplex path (Lectures 2–3)
- unconditional bound: $\mathrm{SLC}_1(x_i^{\mathrm{m}}, g_0) \leq \binom{n}{m} + 1$



Interior point methods match the geometry of the path

Theorem (ADLNV '22)

There is a path-following IPM (the $\textsc{Trust-Region}$ IPM *of Lecture 5) which, started near the central path at gap* g_0*, computes an exact optimal primal-dual pair within*

$$O\Big(\min_{\gamma \in (0,1]} \sqrt{n} \log\left(\tfrac{n}{\gamma}\right) \sum_{i=1}^{n} \mathrm{SLC}_\gamma\left(x_i^{\mathrm{m}}, g_0\right)\Big) \quad \textit{iterations.}$$

- with the Proposition: SLC is the floor for every path-following method and, up to $\tilde{O}(n^{1.5})$, the ceiling for the best one
- an engine for algorithms: the strongly polynomial solver for LPs with ≤ 2 nonzeros per column [DKNOV '24] builds on this theorem

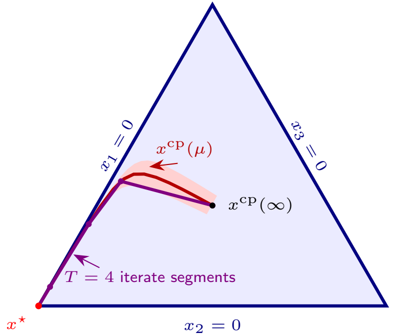
The rest of the lecture computes this quantity.

A benign example: n long straight stretches

For $0 < \varepsilon \ll 1$:

$$\min \sum_{i=1}^{n} \varepsilon^{i-1} x_i \text{ s.t. } \sum_{i=1}^{n} x_i = 1, \quad x \geq \mathbf{0}.$$

- the central path descends through n long, nearly straight segments, visiting the neighborhoods of the vertices $e_n, e_{n-1}, \ldots, e_1$ in order
- yet each x_i^{m} has ≤ 2 pieces ($x_i^{\mathrm{m}} = \min\{1, g/g_i\}$ for thresholds g_i), so $\sum_i \mathrm{SLC}_1(x_i^{\mathrm{m}}) = O(n)$



$n = 3$, $\varepsilon = \frac{1}{50}$: the tube picture from before

Where we are

Block 1 — the SLC framework:

- sandwich: $\frac{1}{n} x_i^{\mathrm{m}} \le x_i^{\mathrm{cp}} \le x_i^{\mathrm{m}}$ coordinatewise
- $\max_i \mathrm{SLC}_{\Omega(1/n)}(x_i^{\mathrm{m}}, g_0)$ lower-bounds every path-following IPM
- some path-following IPM matches $\sum_i \mathrm{SLC}$ up to $\tilde{O}(n^{1.5})$ factors [ADLNV '22]

Block 2 — computing the SLC:

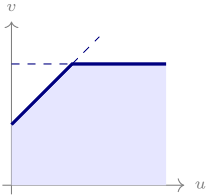
1. can it be large?
   Yes: $2^{n-1} - 1$ on explicit instances, via *tropical geometry* [ABGJ '18]
2. what keeps it small?
   Circuit Imbalance Bound: $\mathrm{SLC}_{1/(n\kappa_A)} \le O(\min\{m, n-m\} \log \kappa_A)$ [DKNOV '24]

Outline

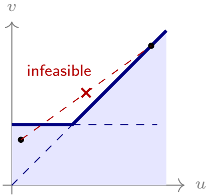
The tropical program

The construction starts from a tropical linear program, in variables $y_1, \ldots, y_{2n+1}$:

$$
\begin{array}{rll}
\min & y_1 & \\
\text{s.t.} & y_1 \leq 2, \quad y_2 \leq 1 & \\
& y_{2j+1} \leq 1 + \min\{y_{2j-1},\, y_{2j}\}, & 1 \leq j \leq n \\
& y_{2j+2} \leq 1 - \frac{1}{2^j} + \max\{y_{2j-1},\, y_{2j}\}, & 1 \leq j < n
\end{array}
$$



$v \leq 1 + \min\{u, 1\}$: intersection — convex



$v \leq \max\{u, 1\}$: union of half-planes — nonconvex

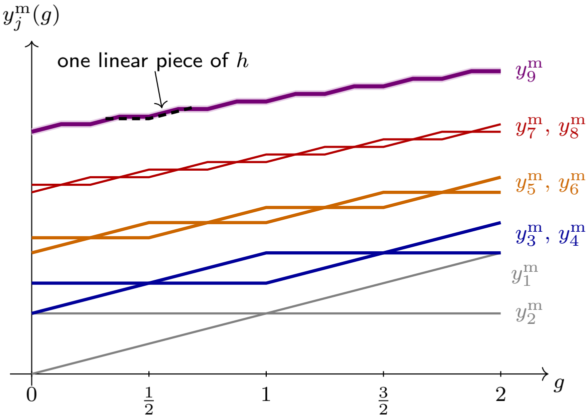
The tropical max central path

$$\min\ y_1 \ \text{s.t.}\ y_1 \leq 2,\quad y_2 \leq 1,$$
$$y_{2j+1} \leq 1 + \min\{y_{2j-1}, y_{2j}\}, \qquad y_{2j+2} \leq 1 - \tfrac{1}{2^j} + \max\{y_{2j-1}, y_{2j}\}$$

$$y_j^{\mathrm{m}}(g) \ := \ \max\{y_j : y \text{ feasible},\ y_1 \leq g\}, \qquad g \in [0,2]$$

- one curve per variable, as in the linear setting — but the feasible region is nonconvex (previous slide), so the curves need not be concave
- the recursion unrolls exactly: $y_1^{\mathrm{m}} = g$, $y_2^{\mathrm{m}} = 1$, then $\min$ / $\max$ of the previous pair, plus the offset
- the shrinking offsets $1/2^j$ alternate *which* predecessor is better as g varies: pieces double at every level

The staircase cascade ($n = 4$)



Level j: 2^j pieces each. The top coordinate y_{2n+1}^{m} is a staircase with 2^n pieces: 2^{n-1} rises of height 2^{1-n}, alternating with flats.

From tropical to linear: the hard instance

Lift $y_j \mapsto x_j := t^{y_j}$ ($t > 1$ large): constants become factors, $\max$ becomes a sum, by the softmax inequalities $\max\{a, b\} \leq \log_t(t^a + t^b) \leq \max\{a, b\} + \log_t 2$.

$$
\begin{array}{lll}
\min & x_1 & \mathrm{LP}(t): \\
\text{s.t.} & x_1 \leq t^2, \quad x_2 \leq t & \\
 & x_{2j+1} \leq t\, x_{2j-1}, \quad x_{2j+1} \leq t\, x_{2j}, & 1 \leq j \leq n \\
 & x_{2j+2} \leq t^{1-1/2^j} \big(x_{2j-1} + x_{2j}\big), & 1 \leq j < n \\
 & x_{2n-1},\ x_{2n},\ x_{2n+1} \geq 0 &
\end{array}
$$

optimal value 0, so the gap of a feasible x is x_1; the $\min$-constraints lift exactly, the $\max$-constraints up to a factor 2

Theorem (ABGJ '18)

For every $\gamma \in (0, 1)$ and $t \geq 2^{\Omega(2^n(n+\log(1/\gamma)))}$: $\quad \mathrm{SLC}_\gamma\big(x^{\mathrm{m}}_{2n+1}\big) \ \geq\ 2^{n-1} - 1.$

Step 1: the collapse

The softmax errors accumulate along the chain — but only linearly:

Collapse estimate (induction through the layers of constraints)

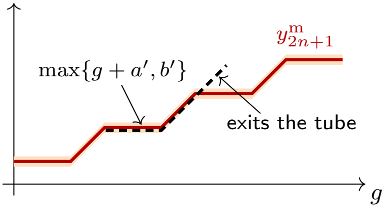
$$\left| \log_t x_j^{\mathrm{m}}(t^{\,g}) \;-\; y_j^{\mathrm{m}}(g) \right| \;\le\; j\,\log_t 2, \qquad g \in [0,2]$$

- for t doubly exponential in n: total error $(2n+1)\log_t 2 \ll 2^{-n}$ — a small fraction of a *single* staircase step
- so on the $\log_t$ scale, the LP max central path is glued to the tropical staircase: it lives in the narrow shaded band of the figure

Step 2: tracking the staircase

Let h be piecewise linear with $\gamma\, x^{\mathrm{m}}_{2n+1} \le h \le x^{\mathrm{m}}_{2n+1}$; set $H(g) := \log_t h(t^g)$.

- by Step 1, H lies in an additive tube of half-width $\delta \ll 2^{-n}$ around the staircase
- a linear piece $h(\xi) = a\xi + b$ (WLOG $a, b \ge 0$) becomes $H(g) \in \max\{g + \log_t a, \log_t b\} + [0, \log_t 2]$: flat, then slope one — a single convex turn
- it can cover one flat and the following rise, but never passes a concave turn



The staircase has $2^{n-1} - 1$ disjoint interior flat–rise pairs, each needing its own piece: $\mathrm{SLC}_\gamma(x^{\mathrm{m}}_{2n+1}) \ge 2^{n-1} - 1.$ [ABGJ '18]

Consequences of the lower bound

Combining with the SLC framework of Block 1:

- no path-following IPM terminates in fewer than $2^{n-1} - 1$ iterations on $\mathrm{LP}(t)$
- in particular, no path-following IPM is strongly polynomial
- by [ADLNV '22], the trust-region IPM needs at most $\tilde{O}(\sqrt{n}\,2^n)$ iterations here: floor and ceiling are *both* exponential — the SLC characterization is tight

The circuit imbalance κ of $\mathrm{LP}(t)$ grows with t — next: bounded κ forces small SLC.

Outline

Circuits and circuit imbalance (Lecture 3 recap)

- circuits of $\mathbf{A}$: support-minimal nonzero vectors of $\ker(\mathbf{A})$, unique up to scaling
- conformal circuit decomposition: every $w \in \ker(\mathbf{A})$ is a sum $w = \sum_{j=1}^{\ell} h^{(j)}$ of $\ell \le n$ circuits with the signs of w — partial sums stay in the box $[\mathbf{0}, w]$
- circuit imbalance: $\kappa_{\mathbf{A}} := \max\left\{\left|h_j/h_k\right| \;:\; h \text{ circuit},\; j,k \in \operatorname{supp}(h)\right\}$
- $\kappa_{\mathbf{A}} = 1$ for network (and all totally unimodular) matrices; Δ-modular integer matrices: $\kappa_{\mathbf{A}} \le \Delta$

The upper bound theorem, and preprocessing

Theorem (DKNOV '24)

With $\kappa := \kappa_{\mathbf{A}}$*:* $\quad \mathrm{SLC}_{1/(n^2\kappa^2)}\big(x_i^{\mathrm{m}}\big) \;\le\; \min\{m,\, n-m\} + 2 \quad$ *for every* $i \in [n]$.

- uniform in g_0, independent of b and c: only the constraint matrix enters
- **preprocessing**: fix a *basic* optimal x^* attaining $x_i^* = x_i^{\mathrm{m}}(0)$ and a basic optimal dual (s^*, y^*); complementary slackness:

$$|\operatorname{supp}(x^*)| \le m, \qquad |\operatorname{supp}(s^*)| \le n-m, \qquad \operatorname{supp}(x^*) \cap \operatorname{supp}(s^*) = \emptyset$$

- gap formula: membership in $\mathcal{P}_g$ reads ${s^*}^{\mathsf{T}} x \le g$

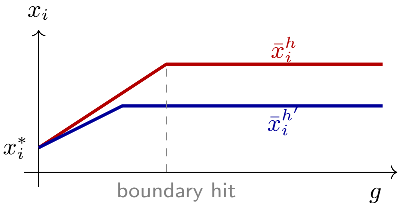
Plan: approximate x_i^{m} by a few elementary curves — one per circuit in a small *cover*.

Circuit curves

For $h \in \ker(\mathbf{A})$, the h-curve $\bar{x}^h(g) := x^* + \alpha^h(g)\, h$ is the farthest point of $\mathcal{P}_g$ reachable from x^* along h:

$$\alpha^h(g) = \min\left(\underbrace{\frac{g}{{s^*}^\mathsf{T} h}}_{\text{gap budget}} \,,\; \underbrace{\min_{j:\, h_j<0} \frac{x_j^*}{|h_j|}}_{\text{first blocking coord.}} \right)$$

$\bar{x}_i^h(g)$ is concave with two pieces: rise from x_i^* at slope $h_i/({s^*}^\mathsf{T} h)$, then constant once the moving point hits the boundary



Circuit curves capture the max central path

Lemma

For every $g \geq 0$ with $x_i^{\mathrm{m}}(g) > x_i^*$ there is a scaled circuit h with $h_i > 0$ and $\bar{x}_i^h(g) \geq x_i^{\mathrm{m}}(g)/n$.

Proof. Let $\hat{x} \in \mathcal{P}_g$ attain $x_i^{\mathrm{m}}(g)$; conformally decompose $\hat{x} - x^* = \sum_{j=1}^{\ell} h^{(j)}$, $\ell \leq n$.

- conformality: each $x^* + h^{(j)}$ stays in the box between x^* and $\hat{x} \Rightarrow$ feasible; optimality of x^* gives $s^{*\mathsf{T}} h^{(j)} = c^\mathsf{T} h^{(j)} \geq 0$, and $s^{*\mathsf{T}} h^{(j)} \leq s^{*\mathsf{T}}(\hat{x} - x^*) \leq g$
- hence $x^* + h^{(j)} \in \mathcal{P}_g$, i.e., $\alpha^{h^{(j)}}(g) \geq 1$; choose k maximizing $h_i^{(k)}$:

$$\bar{x}_i^{h^{(k)}}(g) \;\geq\; x_i^* + h_i^{(k)} \;\geq\; x_i^* + \tfrac{1}{\ell}\textstyle\sum_j h_i^{(j)} \;\geq\; \hat{x}_i/\ell \;\geq\; x_i^{\mathrm{m}}(g)/n \qquad \square$$

Circuit covers

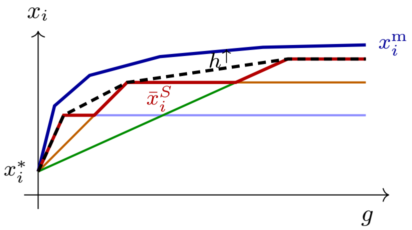
Definition (γ-circuit cover)

$S \subseteq \ker(\mathbf{A})$ (finite, $c^\top h \geq 0$ on S) is a γ-*circuit cover of* i if every scaled circuit h' with $c^\top h' \geq 0$ is dominated: some $h \in S$ has $\bar{x}_i^h \geq \gamma\, \bar{x}_i^{h'}$ pointwise.

Lemma (covers bound SLC)

S a γ-circuit cover of i $\;\Rightarrow\;$ $\mathrm{SLC}_{\gamma/n}(x_i^{\mathrm{m}}) \leq |S| + 1$.

- the upper concave envelope of $|S|$ two-piece curves has $\leq |S| + 1$ pieces
- capture lemma + cover property: the envelope fits in the (γ/n)-tube of x_i^{m}



It remains to construct a small cover — this is where κ enters.

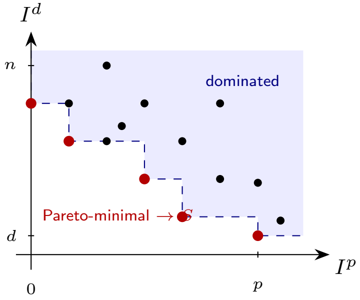
A small cover from the circuit imbalance

Theorem (DKNOV '24)

For every i there is a $(1/(n\kappa^2))$-circuit cover of i of size at most $\min\{m, n-m\}+1$.

Order $x^*_1 \geq \cdots \geq x^*_n$, $s^*_1 \leq \cdots \leq s^*_n$;
$p := |\operatorname{supp}(x^*)| \leq m, \quad \operatorname{supp}(s^*) = \{d, \ldots, n\}$,
$d \geq m+1$.

- **relevant circuits**: $h_i = 1$; discard h with $h_j < 0$, $j \notin \operatorname{supp}(x^*)$; note $1/\kappa \leq |h_j| \leq \kappa$ on $\operatorname{supp}(h)$
- **signature**: $I^p(h) := \max\{j \leq p : h_j < 0\}$ (deepest blocking coord., $\max \emptyset := 0$); $I^d(h) := \max\{j \geq d : h_j > 0\}$ (most expensive paying coord.; never empty, since $x^*_i = x^{\mathrm{m}}_i(0)$)
- $S :=$ one representative per Pareto-minimal signature $\Rightarrow |S| \leq \min\{p+1, n-d+1\}$



Why the cover works: slopes and caps

A two-piece curve is determined by its slope $1/(s^{*\mathsf{T}} h)$ and its cap $\min_{j:h_j<0} x^*_j/|h_j|$ (recall $h_i = 1$). Given a relevant circuit h, take $h' \in S$ with $I^p(h') \le I^p(h)$, $I^d(h') \le I^d(h)$.

- **slopes** ($j' := I^d(h') \le j'' := I^d(h)$; s^* increasing; entries in $[1/\kappa, \kappa]$):

$$s^{*\mathsf{T}} h' \;\le\; n\kappa\, s^*_{j'} \;\le\; n\kappa\, s^*_{j''} \;\le\; n\kappa^2\, s^*_{j''}\, h_{j''} \;\le\; n\kappa^2\, s^{*\mathsf{T}} h$$

  $\Rightarrow$ slope of the h'-curve $\ge \frac{1}{n\kappa^2} \times$ slope of the h-curve

- **caps** (x^* decreasing; the blocking coordinate of h' is *shallower*, hence larger):

$$\mathrm{cap}(h') \;\ge\; \frac{x^*_{I^p(h)}}{\kappa} \;\ge\; \frac{1}{\kappa^2}\,\mathrm{cap}(h)$$

- slope and cap within these factors $\Rightarrow$ pointwise domination up to $1/(n\kappa^2)$ (details: notes)

Consequences: strongly polynomial territory for IPMs

Plugging into [ADLNV '22]: the trust-region IPM solves *any* LP within

$$O\Big(n^{1.5}\,\min\{m, n-m\}\,\log(n\,\kappa_{\mathbf{A}})\Big)\ \text{ iterations}$$

— a bound depending on the constraint matrix alone.

- bounded $\kappa_{\mathbf{A}}$ (network, Δ-modular): strongly polynomial iteration bounds via interior point methods
- caveat: a well-centered start is required, and standard initialization does not preserve κ — a κ-safe initialization needs further ideas [DKNOV '24]
- companion bound: ≤ 2 nonzeros per column $\Rightarrow \mathrm{SLC}_{1/(mn)}(x_i^{\mathrm{m}}) = O(mn)$ — same cover concept, but κ can be unbounded for generalized flow, so the cover construction (over conservative cycles and bicycles) is much more involved

Summary and what comes next

Today: straight line complexity is both floor and ceiling for path-following interior point methods:

- no path-following method beats $\max_i \mathrm{SLC}(x_i^{\mathrm{m}})$ iterations
- explicit instances where this is $2^{n-1} - 1$ [ABGJ '18]: no path-following IPM is strongly polynomial
- $\mathrm{SLC}(x_i^{\mathrm{m}}) \leq \min\{m, n-m\} + 2$ when $\kappa_{\mathbf{A}}$ is bounded [DKNOV '24]

Tomorrow (Lecture 5): the missing piece — the algorithm achieving the matching upper bound: predictor-corrector IPMs, the classical $O(\sqrt{n}\log(1/\varepsilon))$ analysis, and the trust-region step [ADLNV '22].

Exercises: the max central path of maximum flow; greedy and staircase approximation; max central path duality; circuits of generalized flow.